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Hallam Land Management, Taylor Wimpey & Hannick Homes

Eastern Villages, Swindon

BADGER SURVEY REPORT

December 2015

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1.0 INTRODUCTION

1.1 The following report has been produced by FPCR Environment & Design Ltd on behalf of a consortium comprising Hallam Land Management Ltd, Taylor Wimpey and Hannick Homes to provide details of results of habitat surveys undertaken at land to the East of Swindon Wiltshire..

Survey Objectives

1.2 This report is based on a field survey using standard survey methodologies appropriate to badger (Meles meles).

1.3 The objectives of the survey work were to:

- locate all setts and badger activity within the study area
- determine the number of social groups of badger resident within the main study area, their likely ranges, feeding areas, access routes and other relevant parameters to their survival
- provide an evaluation of the likely impact of the proposed development on badgers.

Site Context and Status

1.4 The Study Area is centred on the Ordnance Survey Grid Reference SU 2007 8752 and lies to the east of Swindon close to the village of South Marston.

1.5 The site is primarily used for agriculture at present, with pastoral fields bisected by hedgerow field boundaries in the west, to the south of South Marston, and a range of agricultural uses centred around Rowborough Farm to the east, including arable use, turf production and biomass fuel production.

1.6 The site and wider area to the south has been the focus of a wide range of surveys from at least 1999, including a number completed by FPCR. These surveys included reptile surveys over a number of years both within the site and wider Eastern Villages Allocated site area.

Legislation

1.7 Badgers are protected under the Protection of Badgers Act 1992. This act is based on the need to protect badgers from baiting and deliberate harm or injury. The act makes it an offence to:

- Wilfully kill, injure, take possess or cruelly ill-treat a badger, or attempt to do so
- To intentionally or recklessly interfere with a sett. Sett interference includes disturbing badgers whilst they are occupying a sett, as well as damaging or destroying a sett or obstructing access routes

1.8 A sett is defined as:

"Any structure or place which displays signs indicating current use by a badger".

1.9 Work that disturbs badgers whilst occupying a sett is illegal without a licence, badgers may be disturbed by work near the sett even if there is no direct interference or damage to the sett.

1.10 Licences only allow works to be carried out between July and November inclusive.

2.0 METHODOLOGY

Desktop Study

2.1 The Leicestershire Badger Group and LERC were contacted for badger records locally.

Field surveys

2.2 A walkover survey of the site was carried out on 11th November 2012 and updated in June 2013 and summer and in November 2015 (following leaf-fall).

2.3 Sett locations were noted on a 1:250 scale map. Each sett was examined closely in order to determine the number of entrances and their level of activity.

2.4 Where setts are found, their status and level of activity was noted. Sett status is broadly categorised as follows:

- *Main sett* – usually continuously used with many signs of activity around, a large number of holes and conspicuous spoil mounds
- *Annexe sett* – usually located close to a main sett and connected to it by well used paths. Annexe's may not be continuously occupied
- *Subsidiary sett* – lesser used setts comprising a few holes and without associated well used paths. Subsidiary setts are not continuously occupied
- *Outlier sett* – one or two holes without obvious paths. These are used sporadically

2.5 Level of activity is described as:

- *Well used* – clear of debris, trampled soil mounds and obviously active, with signs of activity such as presence of prints, dislodged guard hairs around the entrances
- *Partially used* – some associated debris or plants at the entrance. Could be used with minimal excavation and usually with signs of activity within the vicinity, for example, badger pathways
- *Disused* – partially or completely blocked entrances

2.6 All of the survey work undertaken was carried out by experienced ecologists. The survey effort and the data recorded for each sett followed the standard methodology used in the two most recent national surveys of badgers (Cresswell et al 1990, Wilson et al 1997).

2.7 In addition to setts, the presence of hairs, footprints, pathways, dung pits and feeding signs were used to plot the patterns of movement of the badgers. Where pathways were confirmed as badger pathways, i.e. there was a clear link to a sett or there was additional evidence of badger activity nearby (such as dung pits, feeding signs or hairs/footprints), these were also noted on the map. If no specific badger activity was recorded, paths were marked as general animal paths (i.e. could be used by deer, fox, as well as badger).

2.8 The 'foraging potential' of areas within the survey area was determined by a subjective assessment of the availability and number of potential food sources. Thus, areas such as short grazed permanent pasture enclosed by a network of dense hedges, which can provide badgers with a variety of opportunities for foraging throughout the year, were classified as 'good potential foraging'. Foraging opportunities in arable or grass-ley fields, woodland or scrub, can be limited

by the season and the management regime, and so were classified as 'moderate potential'. Arable crops and other areas with no food value for badgers were considered 'poor potential foraging'.

3.0 RESULTS

Habitat

3.1 The site is largely in agricultural use, with areas of short grazed and ungrazed grassland and arable land(used for turf production) enclosed by a well-developed network of hedgerows.

Badger walk-over survey

3.2 Two badger setts were identified within the proposed development site in 2013 and a further two setts were identified in 2015. This included a main sett, lower status probable annexe and outlier setts as identified in Figure 1.

3.3 Sett S1 a main sett was located partially within dense hedge and bramble vegetation which made viewing difficult although its extent could be confidently identified (Photograph). A total of at least nine active holes were noted with extensive spoil mounds extending into the abandoned horse paddock to the south and arable field to the north. Evidence of bedding, badger prints, well used scratching posts and hairs confirmed that the site was in active and continuous use.

3.4 A well-used path was also noted to cross the South Marston Stream over a fallen tree where bedding material for use within the sett was clearly being collected.

Photograph 1: Sett 1: entrance holes with fresh spoil.

3.5 Sett 2 was located within a disused stable approximately 100m to the north west of sett S1 and again appeared to be active (photo 2). Hairs and badger paths extending to the northwest were located close by.

Photograph 3: Sett 2 entrance within disused stable.

3.6 An outlier sett (S3) was also identified 180m to the southeast of sett S1 within a hedgerow. A single active hole and two partially active holes, the latter being collapsed holes in the field with signs of rabbit use. Recent digging and bedding within the spoil suggested recent use.

3.7 A further disused outlier sett (S4) was recorded some 600m further south east from S3 on the bank of a wooded ditch. Despite the presence of a large spoil mound no signs of recent use were noted and no further field signs were observed in association with it or adjacent habitats

3.8 Few additional signs of badger were noted elsewhere within the site, although a well-used path was noted extending all of the way to Nightingale Wood to the north east. A mammal path and latrine were identified along the edge of a field in the process of being cut for turf were noted.

4.0 DISCUSSION AND ASSESSMENT OF IMPACTS

4.1 Research indicates that badger territories in lowland England typically range in area from 1km^2 to 5km^2 size being a function of the quality and quantity of foraging habitat present in the area. Where poor foraging opportunity is offered, larger territories are required to support a population.

4.2 Field evidence suggests that the site is occupied by a single badger social group, centred on sett 1. The site was of variable value, with the arable (turf) farmland and taller grassland areas being of least value and woodland, hedgerows and shorter grasslands being of greater value.

Potential Impacts

4.3 Any development impact on the badger population is considered in the following text against the Development Framework Plan. Development proposals comprise the construction of housing and access roads. Extensive areas of Green Infrastructure (GI) will be retained throughout the Site. Further GI corridors will be created through the areas of residential and mixed use development.

Development within 30m of setts (Licensable Actions)

4.4 Initial proposals for the site indicate that the setts will be retained within an extensive area of green infrastructure. Some work is likely to be required around the sett S1 and, in particular, S3 although this could be completed in a sensitive manner and avoiding the more sensitive time of year. Badger setts are however protected from interference or disturbance under the Protection of Badgers Act 1992 and while the setts are likely to be retained, some disturbance could occur during these works. This is unlikely to significantly impact on badgers as it is generally considered that they are relatively tolerant of disturbance, given the fact that they regularly occupy areas where disturbance appears high (as evidenced by their location adjacent to a intensively managed area used for turf production.

Interruption of regular routes of movement through introduction of new roads

4.5 New roads are likely to interrupt corridors of movement along hedgerows within the development site area. The majority are likely to be easily traversed by badgers and given their residential status, unlikely to be heavily used. Nevertheless, given the relatively close proximity of the main sett to the development and severance of important dispersal corridors in the absence of mitigation may occur some impact may occur, particularly where these are lit.

Disruption/loss of foraging habitat

4.6 Construction activities could lead to some loss of suitable foraging habitat arising as a result of the minor loss of improved and species poor semi-improved grassland. However, significant areas of short grassland (optimum foraging habitat will be created adjacent to the main sett. Similarly, extensive short grassland will also be created throughout the green infrastructure thereby providing a potentially important and secure foraging resource.

4.7 In the long-term badgers may also benefit as research suggests that in urban areas, food availability to badgers is considered to be often over-abundant suggesting that the loss in foraging habitat is only likely to be short-term in nature, until garden and sub-urban habitats develop. [1]

5.0 MITIGATION STRATEGY

5.1 Factors key to effective badger mitigation in this case are: -

- Protection of setts during development

[1] Cresswell W.J & Harris S. (1988). Foraging behaviour and home-range utilization in a suburban badger (Meles meles) population. Mammal Review 18, 37-49.

- Where sett closure is essential, ensuring badgers are not harmed during sett closure or construction activities.
- Provision of badger underpass/fencing on the roads where important routes are severed
- Retention/enhancement of foraging habitat and corridors of movement.
- Provision of additional and enhanced badger foraging habitat.

5.2 The results of the badger survey indicate that a single badger social group territory would be affected by the development proposals. Mitigation measures seek to ensure long term sustainability of the group within the area.

5.3 It will be necessary to ensure that newly excavated setts are taken into account as part of the proposals. A full badger survey will therefore be completed for each phase of construction works to assess any change in status of identified sett and any need for further mitigation measures.

5.4 Should any active setts be identified and their retention not possible within the scheme, it may be necessary to close them under licence. Where this is the case closure would be undertaken using one-way badger gates following guidance as to the use, operation and monitoring provided by Natural England. Closure would be undertaken only during the period from July to November (inclusive) unless under exceptional circumstances and licensed by Natural England.

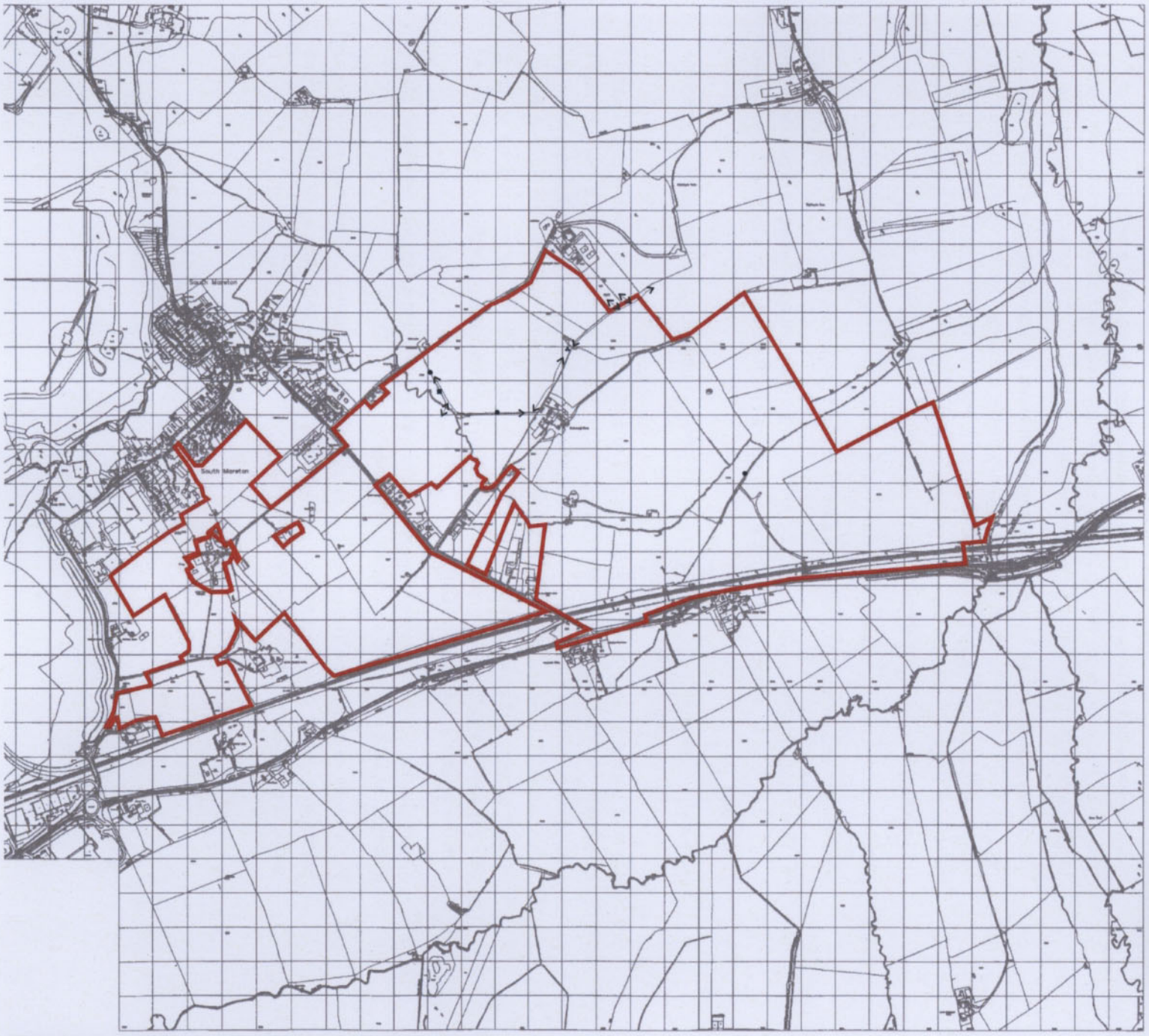
5.5 Other measures to ensure that construction works do not impact on badgers would include:

- Provision and protection buffer strips around retained hedgerows
- Where trenches/voids are created and left overnight, they will be covered or measures taken to ensure that badgers can escape if inadvertently trapped (e.g. placement of scaffold boards).
- Careful consideration given to the location of topsoil storage mounds that can readily become used by badgers for the creation of new setts.

5.6 Corridor habitat is likely to be severed where during the construction of roads occurs. Badger underpass could be incorporated into the road design in close proximity to the existing corridors and at no more than 500m apart. Their design would be based on those contained within the Design Manual for Roads and Bridges (DMRB) and their locations dependant on detailed design. Badger fencing and landscape planting comprising native species will be designed so as to funnel badgers into these features where necessary. Again fencing specification would accord with those recommended in the DMRB.

5.7 Green Corridors will be created throughout the main development, around residential blocks linking to existing habitats in and outside of the site.

5.8 As a result of the above measures the overall impact on badgers is likely to be of minor significance with a potential beneficial impact expected in the long term when residential and public open space habitat becomes established.



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Site Boundary

Main sett

Annexe/Outlier sett

Badger latrine

Mammal path

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Hallam Land Management, Taylor Wimpey & Hannick Homes

Swindon

BADGER SURVEY PLAN

nts @ A3 PH / ph 23.12.2015

Figure 1

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masterplanning environmental assessment landscape design urban design ecology architecture arboriculture